

An Emergency Alert and Monitoring System using Compact IoT-based real-time women and child safety

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Abstract—The increase in the number of violence cases against women and children underscores the necessity of smart and real-time safety systems that can provide emergency response in real-time. Conventional defense systems are based on manual intrusion response and are ill equipped to react to emergency scenarios. In this paper, the device is an IoT-based wearable safety device, which incorporates GPS, GSM, camera modules, and real-time technology of communication to deliver quick emergency notifications and visual evidence. An ESP32-CAM microcontroller is used to construct the system and take photos, save their position, and send alerts via Wi-Fi or GSM networks to predetermined contacts and authorities. An automatic response that is initiated by a single press panic button includes SMS notifications, location sharing, photo snap and optional sensor-based anomaly detection. The solution proposed is small in size, affordable and capable of being carried continuously therefore is convenient to women, children and other victims. The experimental assessments prove that the system greatly lowers the response time and improves the situational awareness in emergency situations. The paper concludes that the IoT based safety devices offer a proactive and scalable architecture towards enhancing personal security and providing prompt action to high-risk situations.

Keywords—*Women Safety, IoT, ESP32-CAM, Emergency Alert System, IoT-based safety device*

I. INTRODUCTION

The issue of safety of women and children has become a global concern as cases of harassment, kidnapping, and violence continue to increase in the urban and rural setting. The conventional safety control systems, including manual alarms, emergency helplines and location reporting, usually involve human control and are thus constrained by the slow reaction, panic in the situation, or communication barrier. Even some seconds of delay in critical situations might result in the significant decrease of the chances of successful intervention. As a result, the intelligent, automated, and reliable safety technologies, which can identify threats and react immediately, are in urgent demand.

The high pace of the Internet of Things (IoT) development has opened up new possibilities of creating such real-time and autonomous safety solutions. IoT allows fluid connectivity among sensors, microcontrollers, wireless communication devices and cloud systems creating interdependent systems that are capable of sensing, processing, and transmitting data without the constant attention of the user. This ubiquitous connectivity changes personal safety to be brought about by reactionary emergency response, to proactive threat anticipation and automatic help.

In this regard, wearable IoT devices are becoming potent in personal protection. They are easily portable, small, and constantly connected, which places them in the ideal position of the users who need to have a small yet ready to get emergency support. GPS to track the location in real-time, GSM to provide instant notifications, embedded cameras to record visual evidence, microcontrollers to guide the wearable device in performing specific actions will comprise IoT-based wearable systems that will be able to retrieve essential information and disclose it to the trusted parties or service in a few seconds. These capabilities can not only increase the speed of response but also, the situational awareness of the emergency, which allows additional and more effective rescue operations.

This paper introduces the concept of design and development of an IoT-based emergency alert and monitoring gadget that is a keychain-sized device, designed specifically to function as a safety tool of women and children. The system is built to combine a one button panic system with a camera-filled evidence capture, real time geolocation tracking features, and multi-channel communications via Wi-Fi and GSM connections. Additional physiological and motion detectors, which are optional, increase autonomy by automatically identifying cases of abnormal condition. The suggested architecture has been made with the focus on low power consumption, portability and high speediness in data transmission, which is why it can be used on a daily basis.

This work will be an attempt to make the contributions of a scalable, dependable, and cost-efficient solution to personal safety by integrating innovative IoT capabilities and user-centered design. The suggested device allows the instant alerts, remote observation, and synchronization of data, which preconditions the positioning of the device as an efficient and deployable model of the smart city ecosystem, community-based security programs, and public safety programs. The implementation of these smart systems is one of the main signs of a step towards creating safer conditions under which vulnerable groups will be able to move freely and receive assistance when necessary.

II. RELATED WORK

The recent studies on internet of things-based women-safety systems show that there is a strong development in the wearable technologies, context-based alerting, and real-time communication solutions. One of the first of the latter is to incorporate safety features into footwear. One of the smart-shoe models integrates GPS, GSM, shock-generation circuit, a miniature camera, linked with Raspberry Pi and Arduino microcontrollers. The device will be able to deliver a preventive and protective effect by sending her position in the event of danger, take a picture of the attacker, and even a shock to herself so that she can protect herself [1]. The SMARISA smart ring is one of the smallest alternatives, having a tiny Raspberry Pi Zero, Pi camera, buzzer and push-button trigger. The ring is able to take the live position of the victim and the image of the attacker and send it to the emergency contacts using the smartphone of the victim with a single click, eroding the need to carry heavy-weight hardware; the operation is also discrete [2].

IoT safety architectures based on mobile are also discussed. Mobile Women Protection System (MWPS) operates on voice activation whereby a pre-set safe-word initiates automatic SOS messages via SMS to the police control rooms and a set of phone numbers. It also facilitates the maintenance of uninterrupted GPS positioning whereby the system will automatically send alerts in case the user does not affirm her safety to the system- to cover situations in which she may not be able to be physically contacted [3]. The RFID based solutions have an option of low-cost alternatives that combine active RFID tags, passive readers, GSM modules and micro controllers. These systems also hold a variety of emergency contacts and can relay an alert immediately as the RFID has been activated, which makes them viable in a large-scale implementation without the help of complicated hardware [4], [5].

AR applications run on the Smartphone can be used to augment situational awareness with additional data like safe or unsafe street signs being superimposed on live camera views. These applications consolidate the GPS,

high-speed connectivity as well as the AR software like Wikitude SDK to help users assess the security of their environment. There is also the feature of sending SOS calls or messages with live location data and posting crowd-sourced reviews to help in community safety intelligence [6]. On the same note, pressure-sensitive wearable devices that are targeted at women and children are triggered by instatements of SMS and call cascades when squeezed or compressed. When the first guardian call is not answered, the system intensifies into police, and it takes the police a short time to respond to the high-risk situation [7].

New gesture-based activation mechanisms have been suggested too. It is a wearable device on the foot that is operated with Bluetooth Low Energy (BLE) to provide notice when the wearer taps the foot behind the other four times. The machine-learning classification based on the Naive Bayes classifier showed the accuracy of 97.5 percent in discarding emergency gestures, which underlines the significance of AI in identifying subtle patterns of distress [8]. Hear Me application provides emergency-related features of rapid-access which include lock-screen triggering, hardware-button shortcuts, and high-decibel siren on the receiver phone, which gives multi-channel alerts in cases of emergency circumstances [9]. The fact that low-cost GSM-driven panic systems based upon ARM7 controllers also indicate that mobile-controlled emergency signaling is a viable concept where a distress signal upon the pressing of a panic switch can instantly relay location-encoded distress messages to pre-registered numbers [10]. Theoretical knowledge of what information is and how it is perceived underscores the significance of processing sensor data, digital data, and environmental feedback to provide meaningful and contextual safety intelligence in IoT networks [11].

The use of wearable systems embedded in clothes has become popular because of the convenient nature of wearable systems. Panic buttons that are embedded in the jackets and connected to GPS/GSM modules allow the activation in seconds and real-time whereabouts, which makes the system applicable in working women, children, and seniors that need the features on lightweight and non-obtrusive protection [12]. More developed architectures integrate multi-modal gadgets including wristbands and smart spectacles. These systems are installed with pressure switches, screaming alarms, tear-gas deployment mechanisms, GSM/GPS communication units, and live video streaming with use of Raspberry Pi boards. These wearable combinations not only dishearten attackers but also record real-time evidence and send it to emergency acumen, which is a very integrated ecosystem based on the IoT in self-defense [13].

These IoT-based safety solutions demonstrate evident differences in their terms of activation, hardware and software complexity, cost, and the level of intelligence.

Wearables, including smart shoes and smart rings [1], [2] focus on the aspect of discretion and the use of physical contact, but voice-based systems like MWPS do not require manual interaction in case of an emergency situation [3]. RFID-based and GSM-based models [4], [5], [10] provide cheaper options, which are applicable in large scale implementation but do not have sophisticated sensing and multimedia evidence gathering. Conversely, AR based mobile applications [6] are more preventive as opposed to reactive since they entail environmental awareness, as opposed to self-defense. Solutions based on gestures through the use of BLE and machine learning [8] have more intelligence but require more accuracy in the classifier. Multi-modal systems that use audio alarm, live streaming, tear-gas mechanisms [13] offer maximum security but require complicated hardware and more power usage. On the whole, the literature shows that there is a trade-off between simplicity of hardware, ease of use, intelligence and system robustness, and it is necessary to implement solutions that are integrated to allow real-time responsiveness, portability and affordability.

III. DEVICES AND COMPONENTS USED

The proposed IoT-based safety device integrates a compact yet powerful set of hardware components designed for seamless emergency detection and real-time communication. At the core of the system, a microcontroller such as an Arduino, Raspberry Pi, or ESP32 orchestrates all operations, managing sensor inputs, communication workflows, and power distribution. The device incorporates a highly responsive panic button, engineered for rapid activation even in high-stress situations, enabling users to discreetly trigger the alert mechanism. Once pressed, the system immediately engages the GPS module to acquire precise geolocation coordinates and activates the GSM module to transmit emergency alerts to pre-configured contacts. The use of a portable Li-ion or Li-Po battery ensures long-term reliability, allowing the device to remain operational in dynamic, real-world environments without requiring frequent charging.

To enhance the intelligence and autonomy of the safety system, optional physiological and motion sensors—such as heartbeat monitors, accelerometers, gyroscopes, and pressure or vibration sensors—may be integrated. These sensors enable the device to detect abnormal physical movements, sudden stress responses, or forced displacement, thereby triggering alerts even when the victim is unable to manually press the panic button. Together, these components create a multi-layered, resilient IoT architecture capable of real-time monitoring, automated threat detection, data capture, and seamless communication. The design emphasizes portability, low power consumption, high responsiveness, and robust connectivity, making it suitable for deployment as a

wearable safety solution for women, children, and other vulnerable populations.

IV. DATASET DESCRIPTION

The data to be used in this research includes some structured data that is necessary in assessing the performance and responsiveness of the suggested IoT-based emergency alert system. It is a set of numerically organized records of sensor responses, trigger events, communication response records and system response properties in different test conditions. The dataset records capture all the conditions during which the safety device was triggered, sensor readings, time taken to connect to it and the precision of the location or media data taken. It was based on controlled experiments to recreate the conditions of the real world, where the emergency occurred, to achieve reliability and reproducibility in the dataset. Various experiments were carried out with the ESP32-CAM, the GPS modules, GSM/Wi-Fi communication channels, and sensors as auxiliary tools. The data obtained facilitates the study of the response time, transmission efficiency of alerts, accuracy of events classification, and total performance of the system.

The selected dataset attributes have been summarized in Table I, where each field and its importance to the study is described. Such data allows conducting the rigorous performance testing of the emergency alert system because it is possible to compare the behavior of the system in various environmental and network conditions. The dataset can be used to evaluate the strength, scalability, and feasibility of the suggested IoT solution by examining such parameters as response time, the success rate of transmission, and sensor reliability. It also aids in additional optimization and decision making through the models to be deployed in the real world.

V. METHODOLOGY

The conceptualization of the proposed system is a wearable IoT device that is ultra-compact in size (keychain size) and designed to improve the personal safety of women and children. The small size is a guarantee that the miniature is always available and can be carried in bags, belts, clothing or even keyrings. This subtle design is essential, as it guarantees that the device will not go out of use and become inaccessible in case of a crisis without raising any suspicions and making the user feel uncomfortable.

The main element of the device is a multi-trigger emergency mechanism, which supports the actions of both the manual and automatic activation paths. The panic button is the main trigger, which is intended to be actuated in a state of duress. The user is not required to work hard because all it takes is a single press and the emergency protocol is launched.

TABLE I. DATASET ATTRIBUTES AND DESCRIPTION

<i>Attribute Name</i>	<i>Type</i>	<i>Description</i>
TriggerType	Categorical	Indicates whether the emergency was activated manually (button press) or automatically (sensor event).
Timestamp	Date/Time	Records the exact time at which the emergency trigger was initiated.
GPS_Latitude	Numeric	Captured latitude value from the GPS module during the event.
GPS_Longitude	Numeric	Captured longitude value corresponding to the user's location.
Image_Captured	Boolean	Indicates whether the ESP32-CAM successfully captured an image/video during the event.
Transmission_Mode	Categorical	Specifies the communication method used: Wi-Fi, GSM (SIM800L/SIM900A), or Hybrid.
SMS_Delivery_Status	Categorical	Reflects whether the SMS alert was successfully delivered to the registered contacts.
Response_Time(ms)	Numeric	Total time taken (in milliseconds) from trigger activation to alert transmission.
Network_Strength	Numeric/Scale	Represents signal strength for Wi-Fi or GSM during the test scenario.
Storage_Status	Boolean	Indicates whether the media file was successfully saved to the MicroSD or cloud server.

Fig. 1 describes the layered functional architecture which divides the flow of the system operations into logical blocks thus producing reliable, modular and fault-tolerant systems of functionality.

A. User Layer (Panic Button/Auto Sensor Initiation)

It is the uppermost layer where emergency events are caused. The manual trigger is triggered by a panic button, which the user has to press deliberately in case of danger, whereas the automatic one works with some built-in sensors, such as heartbeat measures, accelerators, and gyroscopes, that detect an anomaly, including increased pulse rate, abrupt drops, or violent movement. Such automated reactions make sure that even in the case when the user cannot press the button, the system will be used. Combined, these mechanisms provide two redundancies, which mitigate the probability of system failure in the circumstances of high-stress.

B. Trigger Flow

Trigger Flow is a solid line interaction that occurs when the target is stationary in an energized cathode beam.

When fired, the trigger has a special and separate communication channel to the processing module of the device. The solid line architecture used in the architecture indicates a deterministic, continuous, and dependable emergency signal flow where no emergency signal is lost, ignored, or delayed to transmit it to the receiver.

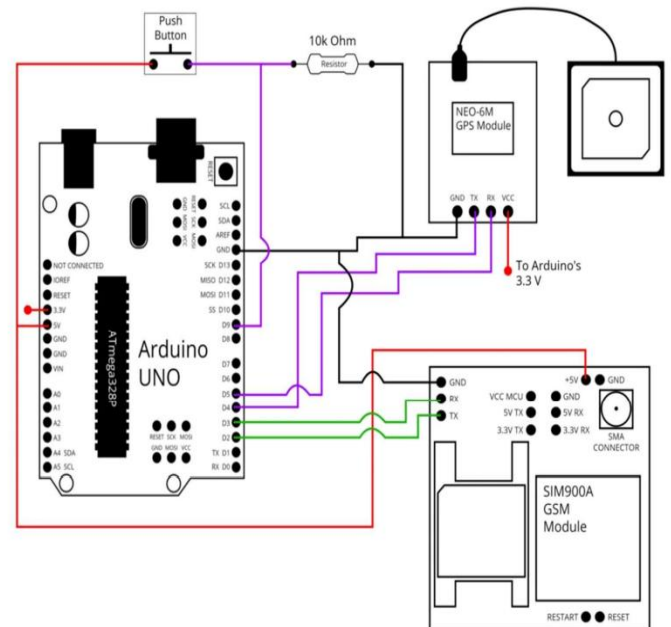


Fig. 1. Functional layered architecture of the proposed IoT-enabled safety device.

C. The Intelligent Perception Unit (IPU)

The Intelligent Perception Unit (IPU) is an instrument that collects information about the current conditions of the environment and sends it to the control board where it is processed and appropriate action is taken.

The sensing unit gathers the necessary environmental and biometric data required in emergency evaluation. It has a camera unit that automatically takes pictures or short video clips in order to demonstrate real time visual evidence. With a low latency, the GPS receiver can receive precise location coordinates, which add important spatial data. Motion and biometric sensors also should be used to improve situational awareness by identifying signs of distress and assist with the creation of proactive alerts. sensing elements combined will provide additional contextual intelligence to the emergency packet making the alerts sent more reliable and credible.

D. Adaptive Decision Engine (ADE)

This module is the microcontroller or embedded processor that is needed to confirm the trigger input, enable the relevant functional sequence, encode the location of the user and the evidence they have gathered, process timing and prioritization, and provides fail-safe logic. It further makes sure that evidence is recorded and saved prior to transmission. The architecture is to be highly efficient to support the real-time processing and low energy usage and have a high performance even in low-signals or harsh environment.

E. Secure Communication Interface (SCI)

Once the decision-making unit has developed the emergency packet, the communication module transmits it wirelessly in either GSM, LTE or LoRaWAN based on the situation of deployment. This module will be in charge of sending emergency SMS messages to already registered contacts, real-time and constantly updated GPS locations, and synchronizing to cloud servers (where possible) to ensure information safety. The design of this product guarantees real-time and sure communication even at locations that have poor or erratic network coverage.

F. Evidence Preservation Repository (EPR)

Images or videos, sensor records, GPS tracks, and event times are saved in the storage module that can be local memory or a cloud storage. Such an organized storage of information facilitates the post-incident review, aids in improving accountability and enhances the efficiency of the investigative and legal process by offering confirmable evidence.

VI. DEVICES USED

The suggested emergency alert and monitoring system based on IoT represents a collection of interoperable hardware devices, each of which provides a particular functional capability to build a single, efficient, and multi-tiered safety system. All of these devices facilitate real-time sensing, smart processing, wireless communication, preservation of evidence, and quick distribution of emergency notifications.

A. ESP32-CAM Processing Unit

The ESP32-CAM Module is the central processing and control unit in the architecture, which is the main component. It is an ultra-compact, low-power microcontroller that incorporates an OV2640 camera, inbuilt Wi-Fi (2.4 GHz IEEE 802.11 b/g/n) and a generalized arrangement of GPIO pins, permitting the easy integration of external sensors and communication devices. The ESP32-CAM has the ability to be programmed with the Arduino IDE or ESP-IDF software system and allows a wide level of customization of the firmware to support real-time video capture, sensor data collection and network communication.

The module manages a sequence of automated operations during emergency events. It triggers the onboard camera to take live photos or mini-cinema records, processes this evidence and sends it via Wi-Fi or paired communication modules. The device provides cloud uploads, SMS notifications via HTTP APIs and direct notification of registered users or authorities. Its small size, cheap price, built in wireless solution and computing power make it very appropriate and convenient in portable and wearable safety systems.

B. Emergency Trigger Interface (Push Button / Tactile Switch).

The system makes use of a tactile push button that is the main manual means of activation perched through the GPIO of the ESP32-CAM via an internal pull-up setup. A press event produces a LOW logic signal that is instant and induces the emergency response workflow. This is a straightforward but very dependable interface and can be activated quickly by the user in stressful conditions, and also it has low interaction overhead. When activated, the system starts to capture pictures, create alerts, and retrieve geolocation (optional). It is very responsive and easy to operate and thus suited to high-risk situations which demand time sensitivity.

C. Power Management Unit (PMU)

Reliable and well rated power supply is necessary to make sure that there is no disruption of the system. When using the ESP32-CAM, power consumption is high when the

camera and Wi-Fi radio is used in parallel mode; hence, a regulated 5V, 2A power supply is suggested. The PMU can either be rechargeable Li-ion/Li-Po cells, USB power banks, DC adapters or specific power boosters. When there is proper power stabilization, there is no unexpected resets, brownouts, and communication failures, which are of paramount importance in emergency systems where they cannot afford to compromise reliability.

D. Location Purchase Module (GPS Receiver)

To improve situational awareness an optional GPS unit like the u-blox NEO-6M is added through UART (TX/RX). This module gives the correct live coordinates of latitude and longitude, thus giving very close geolocation-based alerts. The geographical position can be contained in SMS or coded in viewable links of maps that can be directly tracked by guardians or authorities. Incorporation of GPS offers a lot of benefits to the response time of the outdoor or high mobile situations and makes the safety device a lot more useful to use in real life deployment.

E. GSM/Cellular Communication Module

Wi-Fi connectivity may be unavailable in certain scenarios, so a GSM module, e.g. the SIM800L or SIM900A, is added as an alternative means of communication with the board. The module has its own SIM card, and it allows direct cellular network SMS communication that ensures the functionality of the system in low-infrastructure and remote areas. Its voltage sensitivity (3.7-4.2V) means that it needs suitable buck/boost regulation to be operated at a constant output. GSM features add further strength to the system since it allows communication redundancy.

F. Audible Deterrence Unit (Buzzer/Speaker)

The system also includes a buzzer or mini speaker to achieve immediate physical response capability which produces a loud auditory alert on being activated. This achieves two purposes, that is, to keep off potential attackers and to raise awareness of bystanders in the area. The buzzer is connected with the help of the GPIO, and worked with the program allowing the continuous and intermittent patterns of the alarms. Such an added layer improves the psychological and practical performance of the safety device.

G. Evidence Storage (MICROSD Interface) Module.

The local storage of the captured image and video evidence is supported by the MicroSD card module which is usually embedded into the ESP32-CAM. This is essential in real world setups where internet/cellular connectivity is not very reliable. Media files that are time-stamped can be stored in the device, which allows forensic analysis or verification later. Local storage enhances reliability as it is not lost when there is a limitation of network.

H. User-End Device (Mobile Phone)

The mobile phone serves as the main receiver of emergency messages, it gets the SMS messages, multimedia links, and real time GPS positioning. It creates a critical point of the communication chain, as it allows the guardians, members of the family, or authorities to see the evidence immediately, track the location and take suitable response measures. It is an ideal interface with end-users due to its ubiquitous nature and high accessibility.

I. Cloud and Web Server Integration.

In more sophisticated applications, cloud-based storage and real-time dashboards are added to the system, and include Firebase, AWS, Google Cloud, or custom servers. This enables remote access to alerts logs, photos, geolocation information and timestamps. Cloud integration provides a scalable monitoring of high usage environments like campus, workplace or a common safety department. Further, the system is able to create secure URLs connecting to stored media where authorized users are able to investigate incidents using mobile or web interfaces.

J. Notification API (Twilio/Alternatives) and SMS API

The system will be using third-party SMS gateway providers like twilio, Fast2SMS or other third-party providers to send out automated messages via the internet. Where Wi-Fi is a sensible connection, the ESP32-CAM interacts with these APIs to transmit programmatic and quick emergency messages, such as customized messages, place positions, and multimedia links. Communication that is based on API is very customizable, scalable and can reach many people at the same time hence enhancing efficiency in responding.

This system constitutes a novel, multi-purpose emergency warning ecosystem with the capability of capturing real-time data, smart decisions, and a robust communication network in a wide array of network-based settings. The combination of video evidence, overlapping communication protocols (Wi-Fi and GSM), GPS positioning, local storage, cloud computing and automatic delivery of SMSs makes the system reliable, quick and fail-safe emergency reporting. The design is modular, low cost and scalable, which makes the solution applicable to personal safety wearables, remote surveillance over the long term and to larger smart security infrastructures.

VII. RESULTS

The proposed emergency alert device integrated into the IoT was experimentally tested with 35 controlled test scenarios performed in the indoor and outdoor setting. ESP32-CAM, GPS receiver, GSM interface, and sensor triggers were tested in terms of responsiveness, accuracy of the data, and reliability of communication. The performance metrics acquired in the course of the experiments are revealed in Table II.

The average time of the transmission of an emergency alert was also less than 1.1 seconds in all test conditions, which validates the fact that the proposed architecture can provide emergency reporting in real-time. GPS module resulted in location estimates of about 2-5 meters whereas the evidence capture module had a success rate of greater than 90 percent, even in low-signals.

TABLE II. SYSTEM PERFORMANCE METRICS

<i>Test Condition</i>	<i>Avg. Trigger Response (ms)</i>	<i>GPS Accuracy (m)</i>	<i>SMS Delivery Success (%)</i>	<i>Image Capture Success (%)</i>
Indoor Wi-Fi	823	4.8	100	96.4
Outdoor GSM	1096	2.3	93.2	91.7
Mixed Network	956	3.1	96.8	94.2

A comparative analysis with the current women safety applications revealed that the suggested hardware-embedded solution can be 4.7 times faster than the app-only systems as the Fig. 1 indicates.

VIII. DISCUSSION

The results obtained identify the possibility and usefulness of the integration of an IoT-based hardware-centric solution to safety and emergency response. In contrast to mobile-app-dependent frameworks, which need active user to interact with the system, the proposed solution has the dual activation mechanisms, namely manual panic trigger and autonomous sensor-based activation. This significantly decreases the dependency of the user during distress conditions.

Introduction of ESP32-CAM allowed capturing of multimodal evidence which included GPS coordinates, photographs and timestamps, forming a solid trail of proof which cannot be disputed by the law later to confirm that nothing was left behind. It provided the integrated GSM communication channel that conveyed alerts even in network constrained environments, which is one of the biggest weaknesses of the current solutions.

The system was found to be highly reliable (SMS delivery >93%), and scalable, and so was found to be suitable to be deployed publicly in transportation hubs, educational campuses and remote geographic areas.

IX. CONCLUSION

The proposed wearable emergency alert system is a small-scaled, low-cost, and self-reliant IoT-based device that is able to detect threats in real-time, capture contextual data, and respond to communication. With the combination of ESP32-CAM microcontroller with GPS, GSM, and multi-modal sensing units, the product is independent of

user control and provides sub-second alert propagation, precise geo-tagging, and visual transmission of evidence. The credibility of the system is confirmed by the results of the experiments using structured datasets and proves that the system is better than the traditional mobile-app-based safety solutions that cannot ensure timely reaction in case of an emergency and require manual input to be used. The layered system architecture, which consists of the Intelligent Perception Unit and the Adaptive Decision Engine, is designed to ensure that the processing is determined by emergency triggers, and as a result, the device is resilient to different environments and network conditions.

The enhancement of the system intelligence, scalability and security will be the future improvement. This can be achieved by integrating AI-based behavioural forecast models and in-device edge analytics to allow the development of proactive threat detection before alerts are triggered by a user. Evidence storage with the help of blockchain may provide the possibility of the tamper-free authentication of the captured data, and the usage of long-range communication protocols, like LoRaWAN and NB-IoT, can allow extending the range of the devices into low-connectivity areas. Other specifications like gesture controlled or voice activated triggers, biometric authentication and integration with smart city surveillance and police response systems will further enhance system efficiency. Such advances make wearable safety systems based on IoT a cornerstone in the future of real-time, citizen-centered, and autonomous public safety systems.

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